

Quantum State Tomography

$x_1, \dots, x_N \rightarrow N$ identical systems
in state ρ

Goal $\rightarrow$ Approximate ρ $N^{\frac{1}{2}}$ is large

$\rightarrow$ Qubit Tomography using Pauli Matrices

$$\rho = \frac{1}{2} (\mathbb{I} + \vec{r} \cdot \vec{\sigma})$$

$$\text{Tr}(\sigma_x \rho) = \frac{1}{2} (\text{Tr}(\sigma_x \cdot \rho(\vec{r})))$$

$$= r_x$$

$$\text{Tr}(\sigma_a \rho) = r_a$$

$$\text{Tr}(\rho) = 1 - \sum_{a=0}^3 c_a = 1 \rightarrow \sum_{a \neq b} c_a = 1 - c_b$$

$$p(b) = \frac{1}{2} \left(c_b + \frac{1}{3}(1 - c_b) \right) = \frac{1}{3} \left(c_b + \frac{1}{2} \right)$$

$$c_b = 3p(b) - \frac{1}{2} = 3\text{Tr}(\rho_b \rho) - \frac{1}{2}$$

$$\rho = \sum_{a=0}^3 c_a |\phi_a\rangle\langle\phi_a|$$

$$\tilde{\rho} = \sum_{a=0}^3 \left(\frac{3n_a}{N} - \frac{1}{2} \right) \cdot |\phi_a\rangle\langle\phi_a|$$